

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T48 · Right-Angled Triangles: Pythagoras and Trigonometry

Questions

Student book · questions with writing space

■ 22 pages

FAMILY MAP

Right Angled Triangles Pythagoras Trigonometry

F01	Resolve multi-step two-dimensional right-triangle models	9 Q 35 marks
F02	Use right-triangle trigonometry for lengths	6 Q 18 marks
F03	Use Pythagoras in two dimensions	2 Q 8 marks
F04	Use angles of elevation and depression	4 Q 14 marks

Working Unit 17

4MA1-2H Q10

UNIT 1
5 marks

- 10 ABD and ABC are right-angled triangles.

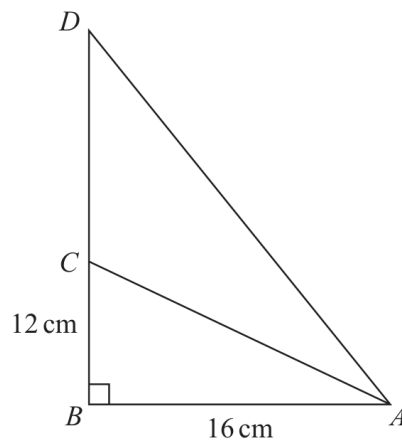


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

Question 07

4MA1-1H Q11

MODULAR UNIT 1

4 marks

11 The diagram shows a roof support.

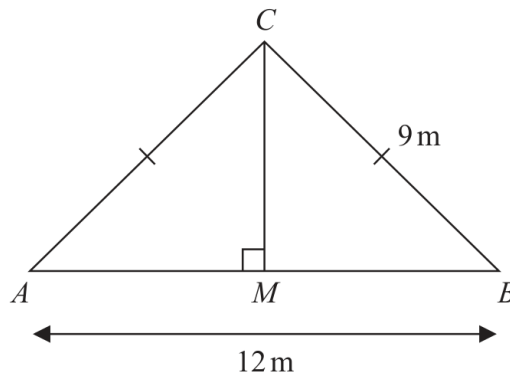


Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$$AC = BC = 9 \text{ m} \quad AB = 12 \text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

Question 19

4MA1-2H Q5

MODULAR UNIT 1

3 marks

- 5 The diagram shows triangle ABC

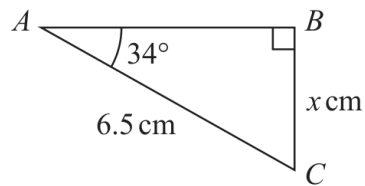


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Question 15

4MA1-2H Q2 whole

MODULAR UNIT 1

4 marks

- 2 The diagram shows triangle ABC and triangle ECD

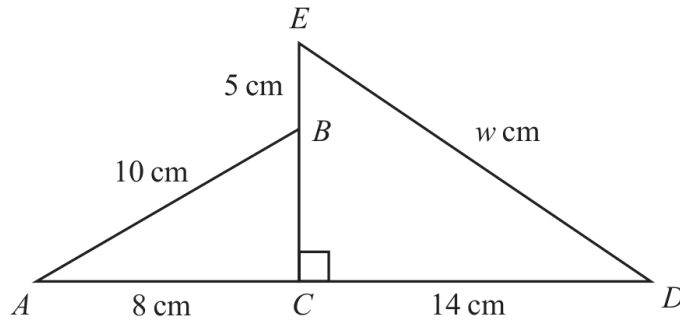


Diagram **NOT**
accurately drawn

ACD and EBC are straight lines.

$$AB = 10 \text{ cm} \quad AC = 8 \text{ cm} \quad EB = 5 \text{ cm} \quad CD = 14 \text{ cm} \quad ED = w \text{ cm}$$

Work out the value of w

Give your answer correct to one decimal place.

$$w = \dots\dots\dots$$

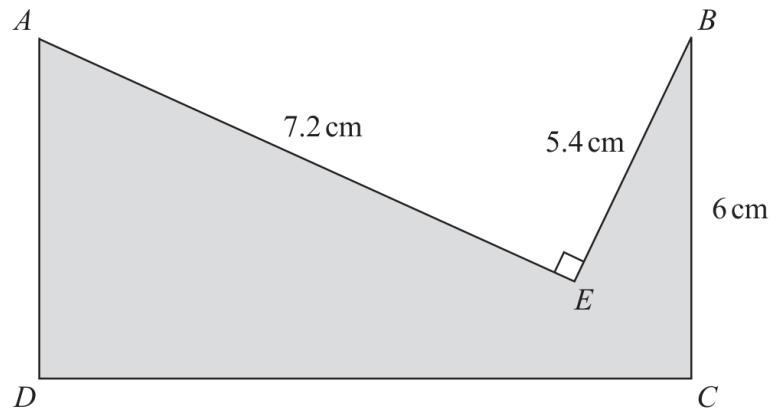
Question 01

4MA1-1H Q5

MODULAR UNIT 1

4 marks

- 5 The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$



$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

Question 17

4MA1-2H Q12

MODULAR UNIT 1

4 marks

12 The diagram shows right-angled triangle ABD

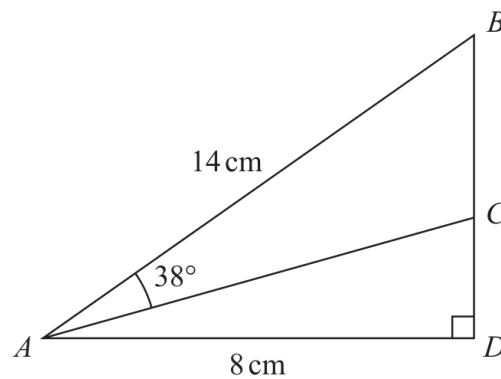


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

Question 09

4MA1-1H Q12 M01

MODULAR UNIT 1

4 marks

- 12 ABC is a right-angled triangle.
 D is a point on BC

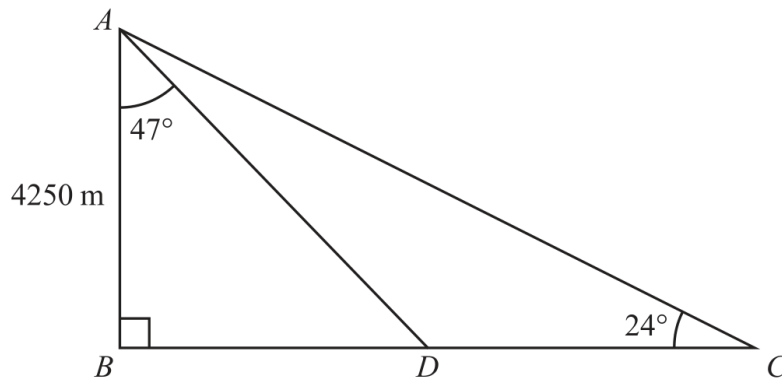


Diagram **NOT**
 accurately drawn

$$AB = 4250 \text{ m} \quad \text{angle } BAD = 47^\circ \quad \text{angle } BCA = 24^\circ$$

Work out the length of DC

Give your answer correct to the nearest integer.

..... m

Question 20

4MA1-2H Q3 M01

MODULAR UNIT 1

3 marks

- 3 The diagram shows triangle PQR

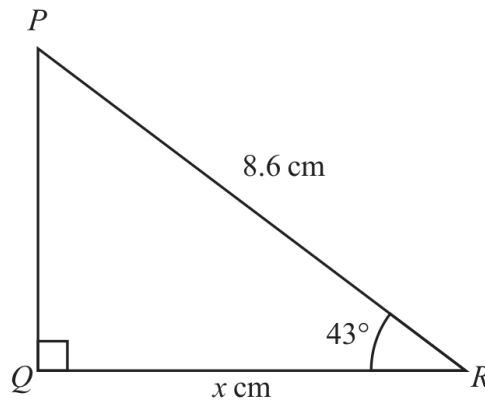


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Question 19

4MA1-2H Q6 Q6

MODULAR UNIT 1

3 marks

- 6 The diagram shows triangle PQR .

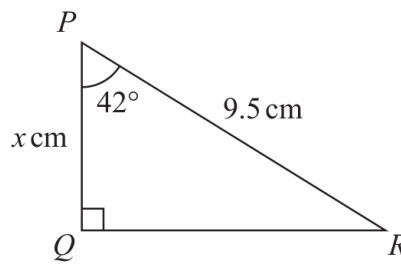


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Question 04

4MA1-1H Q3

MODULAR UNIT 1

3 marks

- 3 The diagram shows triangle ABC

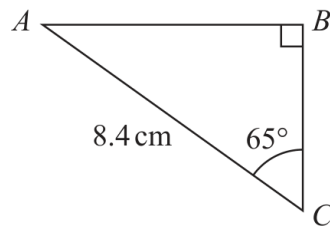


Diagram **NOT**
accurately drawn

Work out the length of the side AB

Give your answer correct to 3 significant figures.

..... cm

Question 24

4MA1-2H Q14

MODULAR UNIT 1

3 marks

- 14 A zip wire is shown as the dashed line AC in the diagram.

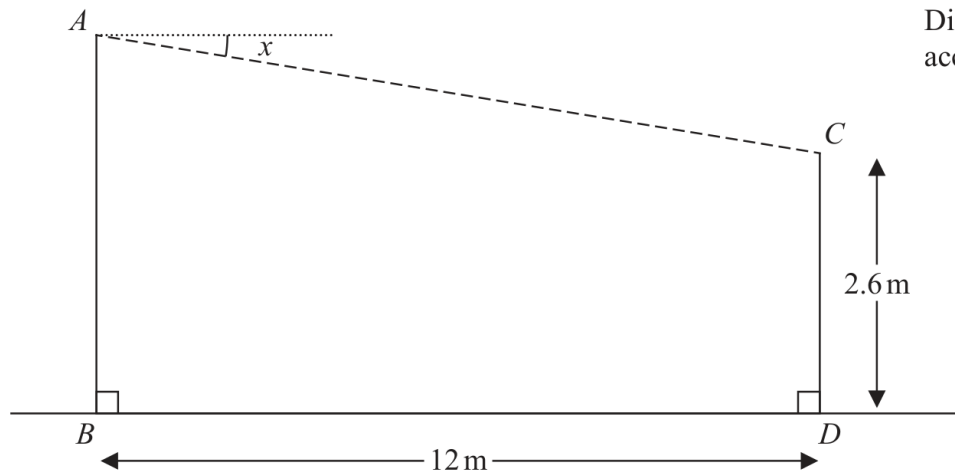


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

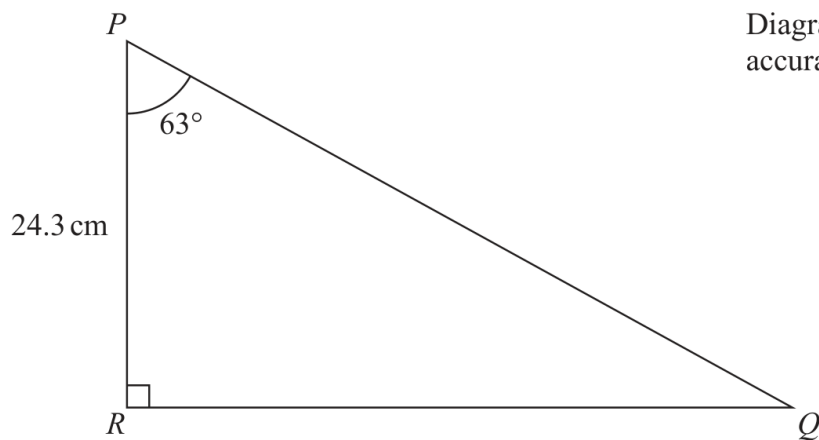
$$CD = 2.6 \text{ m} \quad BD = 12 \text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

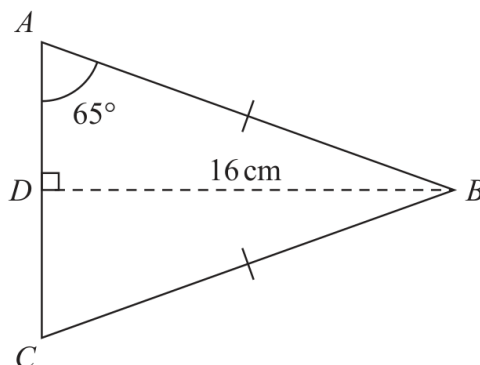
..... m

Diagram **NOT**
accurately drawn



..... cm

Diagram **NOT**
accurately drawn

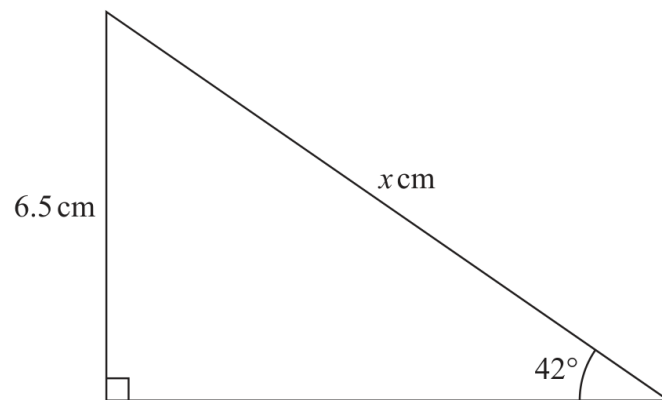


Angle $DAB = 65^\circ$

Work out the perimeter of triangle ABC .
Give your answer correct to one decimal place.

..... cm

Diagram **NOT**
accurately drawn



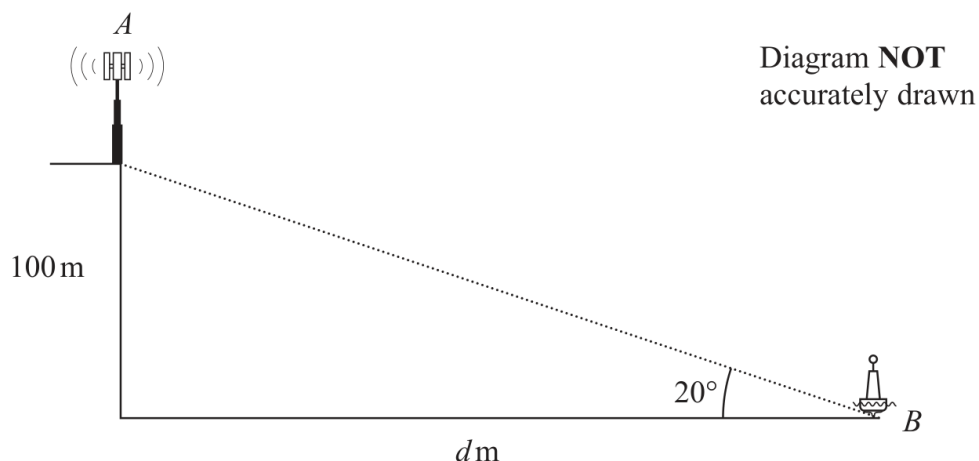
Work out the value of x .
Give your answer correct to one decimal place.

A right-angled triangle is shown. The horizontal side (base) is labeled 30 cm. The vertical side (height) is labeled h cm. The hypotenuse is labeled 52 cm. A right-angle symbol is at the top vertex.

Diagram **NOT**
accurately drawn

Calculate the value of h .
Give your answer correct to 3 significant figures.

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

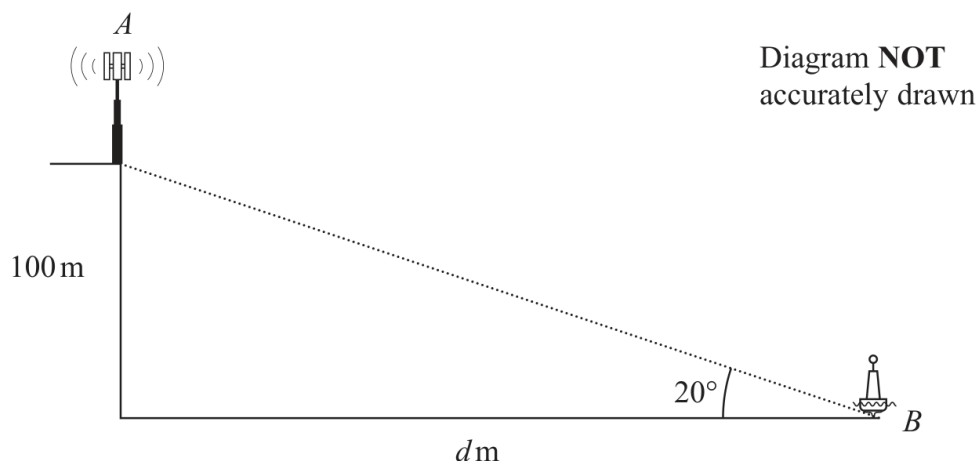
The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



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- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

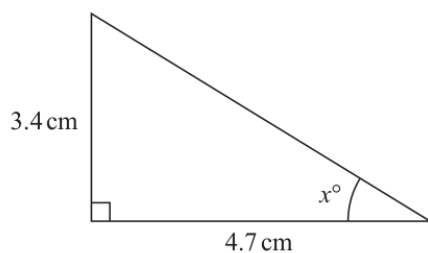
The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

Diagram **NOT**
accurately drawn



Calculate the value of x .
Give your answer correct to one decimal place.

- 10 The diagram shows an isosceles triangle, with base length 24 cm.

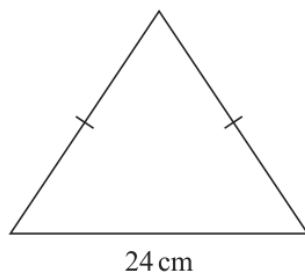


Diagram **NOT**
accurately drawn

The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm²

Diagram of a right-angled triangle ABC with the right angle at vertex B . The base BC is 15 cm . A point D is located on the vertical side AB such that the segment DB is 9 cm . A line segment DC is drawn, and the hypotenuse AC is labeled as 21 cm .

Work out the size of angle ACD
Give your answer correct to one decimal place.

- 11 The diagram shows a right-angled triangle ABC

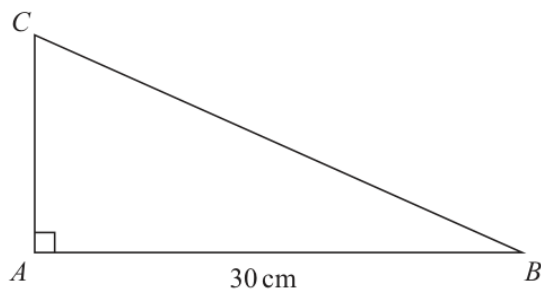


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm